ENGINEERS WALLAH

CHEMISTRY

Lecture VII

Laws of Chemical Combination and Mole Concept

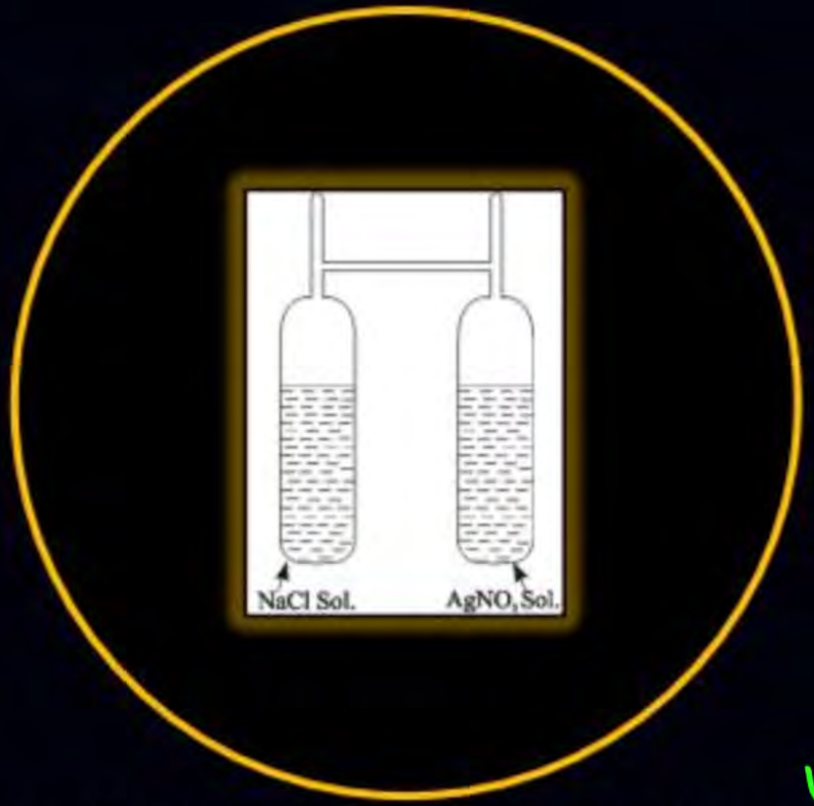


By: Sunil Bhaiya

TOPICS *to be covered*

1. Laws of Chemical Combination
2. Mole Concept

Knowledge Ride On



Laws of Chemical Combination

Knowledge Ride On



Mole Concept

Knowledge Ride On



Efficiency Hacks



Glossary

- Laws of chemical combination (रासायनिक संयोजन के नियम)
- Law of conservation of mass (द्रव्यमान के संरक्षण का नियम)
- Law of definite proportion (निश्चित अनुपात का नियम)
- Atomic mass (परमाणु भार)
- Molecular mass (आण्विक द्रव्यमान)
- Formula unit mass (सूत्र इकाई द्रव्यमान)
- Mole (मोल)
- Molar mass (मोलर द्रव्यमान)
- Reactant (अभिकारक)
- Product (उत्पाद)

Concept Bites – Laws of Chemical Combination

Laws of Chemical Combination

Law of Conservation of Mass
(द्रव्यमान के संरक्षण का नियम)

-Given in → 1789

-Given by → Antoinne Laurent Lavoisier
(Father of modern chemistry)

Law of Definite/Constant Proportion
(निश्चित अनुपात का नियम)

1794 / 1799 (in some sources)

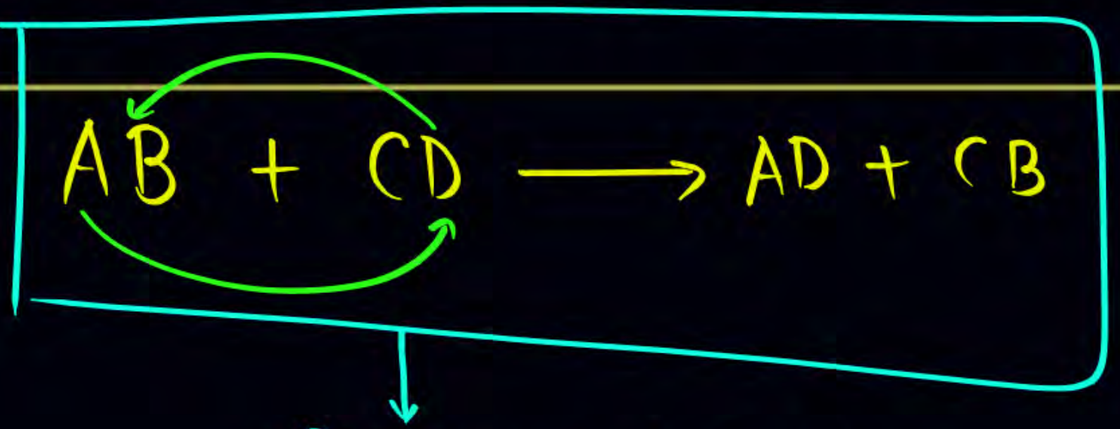
Joseph L. Proust

Law of Conservation of Mass (द्रव्यमान के संरक्षण का नियम)

→ not nuclear rxns

In simple chemical reactions, mass is neither created nor destroyed, i.e. mass of reactants (अभिकारकों) is always equal to mass of products (उत्पादों).

ex:



Rearrangement of atoms

∵ of the rearrangement; no. of atoms are equal on both sides. ↳ $(\text{Mass})_R = (\text{Mass})_P$

example

$$\underset{(56\,g)}{CaO} + \underset{(44\,g)}{CO_2} \longrightarrow \underset{(100\,g)}{CaCO_3}$$

Law of Definite Proportion (निश्चित अनुपात का नियम)

Constant

1 g of hydrogen reacts 8 g of oxygen to form 9 g of water

A specific chemical compound (irrespective of the source) is always made up of the same elements combined together in (same fixed proportion by mass.) → (II)

(H) (H)

H_2O

Source → Tap water, Sea water, Lake water

H_2O

Hydrogen and oxygen element

(II)

	Hydrogen		Oxygen
Volume ratio	2	:	1
Mass ratio	~~2~~ 1	:	~~16~~ 8

1:8

Meaning

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Irrespective of the source, a pure sample of water always contains 88.89% mass of oxygen and 11.11% mass of hydrogen. This is explained by the law of _____.

A. conservation of mass

B. constant proportion

C. multiple proportion

D. constant volume

H_2O

Hydrogen : Oxygen

Volume ratio 2 : 1

Mass ratio 2 : 16

1 : 8

1 g of H_2 reacts with 8 g of O_2 to form 9 g of H_2O

Mass % of hydrogen → $\frac{1}{9} \times 100\%$ → 11.11%

Mass % of oxygen → 100 − 11.11% → 88.89%

Mass % of hydrogen in H_2O $\rightarrow$ $\frac{2}{18_9} \times 100\%$ $\rightarrow$ $\frac{100}{9}\%$ $\rightarrow$ $\underline{11.11\%}$

Question

Which among the following contains the two elements in the same fixed proportion of 1:16 by mass?

A Water

B Hydrogen sulphide

C Carbon dioxide

D Sodium chloride

Hydrogen Oxygen H_2O

Mass ratio 2 : 16

1 : 8

Hydrogen Sulphur H_2S

Mass ratio 2 : 32

1 : 16

Carbon Oxygen CO_2

Mass ratio 12 : 32

3 : 8

Question

Apply the law of constant proportion to calculate the mass of oxygen that will be used for combustion of 5 g of hydrogen gas?

A 8 g

B 16 g

C 40 g

D 20 g

1 g of H_2 requires $\longrightarrow$ 8 g of oxygen

5 g of H_2 -11 $\longrightarrow$ x

$x \times 1 = 8 \times 5$

$x = 40g$

Question

What will be the mass of carbon in 2 g of carbon dioxide?

A .845 g

B .666 g

C 3 g

D .545 g

	Carbon	:	Oxygen
Mass	~~12~~	:	~~32~~
ratio	~~6~~	:	~~16~~
	3	:	8

3 : 8

3 g of C reacts with
8 g of O_2 to form
11 g of CO_2

3 g of C present in → 11 g CO_2

x → 2 g CO_2

$\Rightarrow x \times 11 = 6$

$\Rightarrow x = \frac{6}{11} = .545\ g$

Question

What will be the mass of carbon dioxide produced if 3 g of carbon is burnt in 50 g of oxygen?

A. 22 g

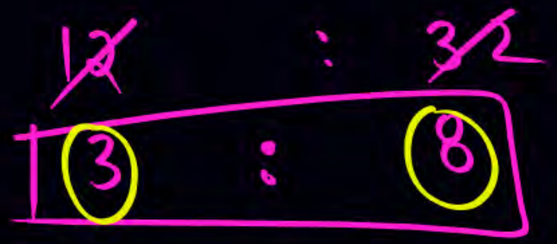
B. 44 g

C. 11 g

D. 32 g

Carbon Oxygen $[CO_2]$

Mass ratio



Even if 50 g is supplied, carbon reacts with 8 g to form 11 g CO_2

Important Fact to Remember

Father of modern atomic theory

In 1808, John Dalton gave an atomic theory to explain the laws of chemical combination.

One of the main postulate of the theory was:

All matter is made up of extremely small indivisible particles called atoms.

But today we know that atom is divisible into sub-atomic particles like electron(e^-), proton(e^+) & neutron(e).

Concept Bites – Mole Concept



Mole Concept

Covalent compounds & elements

Avogadro number/constant

Molar Mass

1 mole atoms/molecules/formula units = 6.02×10^{23} atoms/molecules/formula units = Gram atomic/Gram molecular/Gram formula unit mass

(ionic compounds)

elements

UNITARY METHOD

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

What is the number of moles for 52 g of He?

RRB JE 2019

A 14

B 12

C 13

D 11

Remember the Below Line

Mass of 1 He atom → 4u

1 mole atoms/molecules/ formula units	=	6.02×10^{23} atoms/molecules/formula units	=	Gram atomic/Gram molecular/Gram formula unit mass

1 mole of He = 4 g

x = 52 g

$x \times A\ g = \frac{52}{13}\ g$

$x = 13$

Question

What is the mass of .5 moles of nitrogen atoms?

RRB JE 2019

N

nitrogen molecule → N_2

A 14

B 21

C 7

D 70

Remember the Below Line

Mass of 1 N atom = 14 u

1 mole atoms/molecules/formula units = 6.02×10^{23} atoms/molecules/formula units = Gram atomic/Gram molecular/Gram formula unit mass

1 mole of nitrogen atom = 14 g

.5 mole -11 ——— = x

$x = \frac{1.8 \times 14}{18} \times 7$

$x = 7$

Question

Calculate the number of particles in 46 g of Na atoms.

RRB JE 2019

A 12.044×10^{23}

B 6.022×10^{23}

C $46 \times 6.022 \times 10^{23}$

D 46

Remember the Below Line

Mass of 1 Na atom → 23 u

1 mole atoms/molecules/ formula units	=	6.02×10^{23} atoms/molecules/formula units
=		Gram atomic/Gram molecular/Gram formula unit mass

6.02×10^{23} particles of Na = 23 g

x = 46 g

$x \times 23\,g = 6.02 \times 10^{23} \times 46\,g$

$x = 12.04 \times 10^{23}$

Question

NH_3

A glass jar contains 1.7 g of ammonia gas. Calculate the number of mole present in the gas jar?

RRB JE 2019

A 1 mole

B .01 mole

C .1 mole

D .2 mole

Remember the Below Line

Mass of one molecule of NH_3

$14+(3\times1)=17u$

1 mole atoms/molecules/ formula units	=	6.02×10^{23} atoms/molecules/formula units	=	Gram atomic/Gram molecular/Gram formula unit mass

1 mole of NH_3 = 17 g

x = 1.7 g

$x \times 17\,g = 1.7\,g$

$x = \frac{1.7}{17\times10} = \frac{1}{10} = 1$ mole

Question

valency is 3

A trivalent metal 'M' was made to react with oxygen to yield .25 mole of metal oxide. Which of the following is a correct statement about the product of this reaction? The reaction product contains:

RRB JE 2015

A. 1.506×10^{23} molecules of oxide of formula M_2O_3

B. 1.506×10^{23} molecules of oxide of formula M_3O_2

C. 1.506×105.75 molecules of oxide of formula M_3O_2

D. 1.506×105.75 molecules of oxide of formula M_2O_3

M	O
Valency 3	2

M_2O_3

Remember the Below Line

1 mole atoms/molecules/ formula units	=	6.02×10^{23} atoms/molecules/formula units	=	Gram atomic/Gram molecular/Gram formula unit mass

1 mole M_2O_3 = 6.02×10^{23}

.25 mole M_2O_3 = x

$$x = \frac{3.01}{6.02} \times 10^{23} \times \frac{25}{100} \times \frac{1}{2}$$

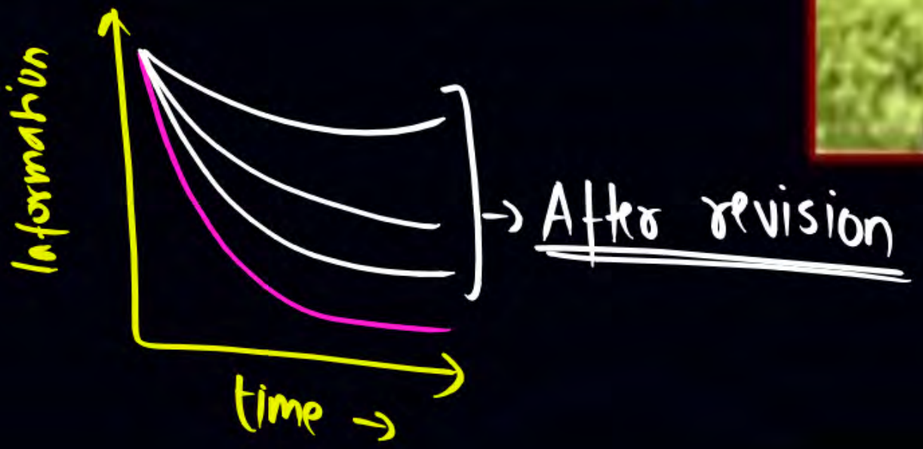
$$x = \frac{3.01}{2} \times 10^{23}$$

$$x = 1.50 \times 10^{23}$$

Efficiency Hacks!

Tips to Score High

- Revise! Revise! Revise!






#sbsathhai
#pwsathhai
Thank
You !